

Technology Stack

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose

1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>Streamlit (Python), HTML, CSS</i>	<i>Provides an interactive and responsive web interface with minimal coding and easy data visualization.</i>
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2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Scikitlearn, Pandas, NumPy</i>	<i>Handles data preprocessing, machine learning model training, and prediction efficiently.</i>
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3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset</i>	Stores HDI dataset and prediction records. CSV is simple for ML projects, while SQLite supports structured storage.
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4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Streamlit Community Cloud</i>	<i>Easy deployment, free for student projects, and accessible from any browser.</i>
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5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Tracks code changes, enables collaboration, and maintains project versions.</i>
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Kaggle Dataset, Matplotlib, Seaborn, Plotly</i>	<i>Kaggle provides the dataset, while visualization libraries generate graphs and charts for HDI analysis.</i>
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